



Part 2: Granting Data Access (The Fix)  
Assign Role- Key Vault Secrets Officer.

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Microsoft Azure

Home > skskeyvault26 | Access control (IAM)

Add role assignment

|                                                |                                                                                                                             |             |          |                      |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------|----------|----------------------|
| App Configuration Reader                       | Grants permission for read operations for App Configuration resources. This role does not grant access to data pla...       | BuiltInRole | None     | <a href="#">View</a> |
| Key Vault Administrator                        | Perform all data plane operations on a key vault and all objects in it, including certificates, keys, and secrets. Canno... | BuiltInRole | Security | <a href="#">View</a> |
| Key Vault Certificate User                     | Read certificate contents. Only works for key vaults that use the 'Azure role-based access control' permission model.       | BuiltInRole | None     | <a href="#">View</a> |
| Key Vault Certificates Officer                 | Perform any action on the certificates of a key vault, except manage permissions. Only works for key vaults that us...      | BuiltInRole | Security | <a href="#">View</a> |
| Key Vault Contributor                          | Lets you manage key vaults, but not access to them.                                                                         | BuiltInRole | Security | <a href="#">View</a> |
| Key Vault Crypto Officer                       | Perform any action on the keys of a key vault, except manage permissions. Only works for key vaults that use the '...       | BuiltInRole | Security | <a href="#">View</a> |
| Key Vault Crypto Service Encryption User       | Read metadata of keys and perform wrap/unwrap operations. Only works for key vaults that use the 'Azure role-ba...          | BuiltInRole | Security | <a href="#">View</a> |
| Key Vault Crypto Service Release User          | Release keys. Only works for key vaults that use the 'Azure role-based access control' permission model.                    | BuiltInRole | None     | <a href="#">View</a> |
| Key Vault Crypto User                          | Perform cryptographic operations using keys. Only works for key vaults that use the 'Azure role-based access contr...       | BuiltInRole | Security | <a href="#">View</a> |
| Key Vault Data Access Administrator            | Manage access to Azure Key Vault by adding or removing role assignments for the Key Vault Administrator, Key Va...          | BuiltInRole | None     | <a href="#">View</a> |
| Key Vault Purge Operator                       | Allows permanent deletion of soft-deleted vaults.                                                                           | BuiltInRole | None     | <a href="#">View</a> |
| Key Vault Reader                               | Read metadata of key vaults and its certificates, keys, and secrets. Cannot read sensitive values such as secret cont...    | BuiltInRole | Security | <a href="#">View</a> |
| Key Vault Secrets Officer                      | Perform any action on the secrets of a key vault, except manage permissions. Only works for key vaults that use th...       | BuiltInRole | Security | <a href="#">View</a> |
| Key Vault Secrets User                         | Read secret contents. Only works for key vaults that use the 'Azure role-based access control' permission model.            | BuiltInRole | Security | <a href="#">View</a> |
| Operator Nexus Key Vault Writer Service Rol... | (Preview) Provides Azure Operator Nexus services the ability to write to a Key Vault. This role is in preview and subj...   | BuiltInRole | None     | <a href="#">View</a> |

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RoleMembersConditionsReview + assign

Role

Key Vault Secrets Officer

Scope

/subscriptions/907eda19-180e-44b4-ba2d-b82d41f8f95e/resourceGroups/rg-demo1/providers/Microsoft.KeyVault/vaults/skskeyvault26

Members

| Name                 | Object ID                            | Type |
|----------------------|--------------------------------------|------|
| Lovely Sharma(Guest) | e558c85a-99a5-47ed-988a-574ade4e6bfe | User |

Description

No description

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... Adding Role assignment

Lovely Sharma(Guest) is being added as Key Vault Secrets Officer for skskeyvault26.

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portal.azure.com/#@lovelysh2604gmail.onmicrosoft.com/resource/subscriptions/907eda19-180e-44b4-ba2d-b82d41f8f95e/resourceGroups/rg-demo1/provider...

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Key vault

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Automation

Check access Role assignments Roles Deny assignments Classic administrators

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View my level of access to this resource.  
View my access

Check access  
Review the level of access a user, group, service principal, or managed identity has to this resource. Learn more  
Check access

Grant access to this resource  
Grant access to resources by assigning a role. Learn more  
Add role assignment

View access to this resource  
View the role assignments that grant access to this and other resources. Learn more  
View

View deny assignments  
View the role assignments that have been denied access to specific actions at this scope. Learn more  
View

Added Role assignment  
Lovely Sharma(Guest) was added as Key Vault Secrets Officer for skskeyvault26.

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### Part 3: The "Permission Gap" Demo

1. Once deployed, go to the resource.
2. On the left menu, under **Objects**, click **Secrets**.

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portal.azure.com/#@lovelysh2604gmail.onmicrosoft.com/resource/subscriptions/907eda19-180e-44b4-ba2d-b82d41f8f95e/resourceGroups/rg-demo1/pr...

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Key vault

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Generate/import Refresh Restore Backup Manage deleted secrets View sample code

The secret 'MySecret' has been successfully created.

| Name     | Type | Status    | Expiration date |
|----------|------|-----------|-----------------|
| MySecret |      | ✓ Enabled |                 |

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## Part 4: Creating a Key (Encryption) and Assign Role- Key Vault Crypto Officer

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### Add role assignment

Role Members Conditions **Review + assign**

**Role** Key Vault Crypto Officer

**Scope** /subscriptions/907eda19-180e-44b4-ba2d-b82d41f8f95e/resourceGroups/rg-demo1/providers/Microsoft.KeyVault/vaults/skskeyvault26

**Members**

| Name                 | Object ID                            | Type |
|----------------------|--------------------------------------|------|
| Lovely Sharma(Guest) | e558c85a-99a5-47ed-988a-574ade4e6bfe | User |

**Description** No description

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### skskeyvault26 | Keys

Key vault

[Generate/import](#) [Refresh](#) [Restore Backup](#) [Manage deleted keys](#)

[Overview](#) [Activity log](#) [Access control \(IAM\)](#) [Tags](#) [Diagnose and solve problems](#) [Access policies](#) [Resource visualizer](#) [Events](#) [Objects](#) [Keys](#) [Secrets](#) [Certificates](#) [Settings](#) [Monitoring](#) [Automation](#)

The key 'Myvaultkey' has been successfully created.

| Name       | Status    | Expiration date |
|------------|-----------|-----------------|
| Myvaultkey | ✓ Enabled |                 |

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## Part 5: The Implementation Code

```

1  using Azure.Identity;
2  using System.Text;
3  using Azure.Security.KeyVault.Keys.Cryptography;
4
5
6  0 references
7  internal class Program
8  {
9      0 references
10     static void Main(string[] args)
11     {
12         string keyVaultUrl = "https://skskeyvault26.vault.azure.net/";
13         string keyName = "Myvaultkey";
14
15         var credential = new DefaultAzureCredential();
16
17         var cryptoClient = new CryptographyClient(
18             new Uri($"{keyVaultUrl}keys/{keyName}"),
19             credential
20         );
21
22         string plainText = "Hello SmartBank!";
23         byte[] data = Encoding.UTF8.GetBytes(plainText);
24
25         // Encrypt
26         var encryptResult = cryptoClient.Encrypt(
27             EncryptionAlgorithm.RsaOaep,
28             data
29         );
30
31         Console.WriteLine($"Encrypted: {Convert.ToBase64String(encryptResult.Ciphertext)}");
32
33         // Decrypt
34         var decryptResult = cryptoClient.Decrypt(

```

## Part 6: Output

```

1  using Azure.Identity;
2  using System.Text;
3  using Azure.Security.KeyVault.Keys.Cryptography;
4
5
6  0 references
7  internal class Program
8  {
9      0 references
10     static void Main(string[] args)
11     {
12         string keyVaultUrl = "https://skskeyvault26.vault.azure.net/";
13         string keyName = "Myvaultkey";
14
15         var credential = new DefaultAzureCredential();
16
17         var cryptoClient = new CryptographyClient(
18             new Uri($"{keyVaultUrl}keys/{keyName}"),
19             credential
20         );
21
22         string plainText = "Hello SmartBank!";
23         byte[] data = Encoding.UTF8.GetBytes(plainText);
24
25         // Encrypt
26         var encryptResult = cryptoClient.Encrypt(
27             EncryptionAlgorithm.RsaOaep,
28             data
29         );
30
31         Console.WriteLine($"Encrypted: {Convert.ToBase64String(encryptResult.Ciphertext)}");
32
33         // Decrypt
34         var decryptResult = cryptoClient.Decrypt(

```

Encrypted: SDCvJngRQmQ3niTPGsk6Rst6hYw/ea07Uvt2gc0U/sxY7/SeavU2Fo68P4pi/IpL72cKxYjP5CbuwoLlY4wDoFwXma94MTT0eY0xahdIb...

Decrypted: Hello SmartBank!